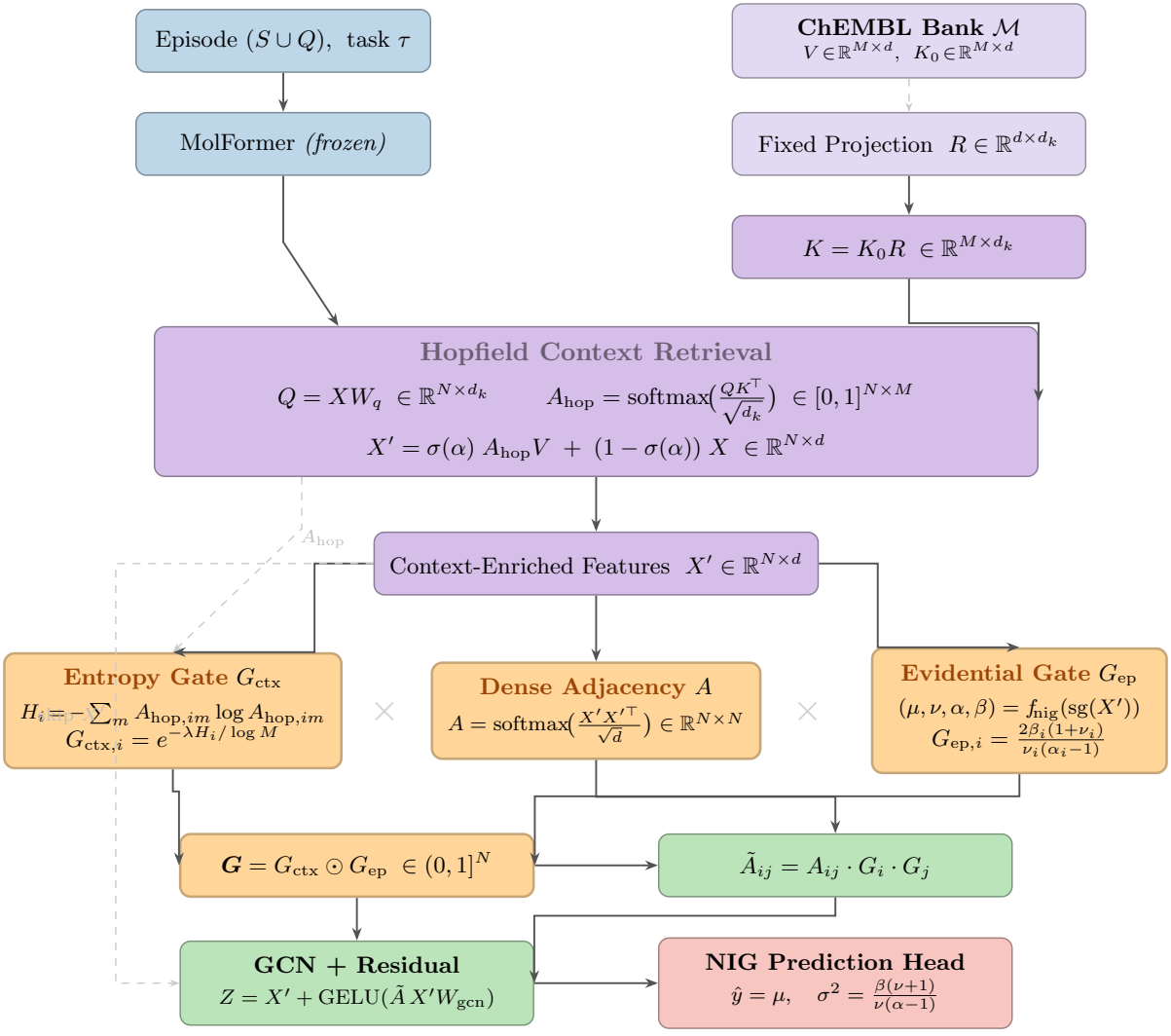
Episode $(S \cup Q)$, task τ
MolFormer *(frozen)*
ChEMBL Bank $\mathcal{M}$
$V \in \mathbb{R}^{M \times d}$, $K_0 \in \mathbb{R}^{M \times d}$
Fixed Projection $R \in \mathbb{R}^{d \times d_k}$
$K = K_0 R \in \mathbb{R}^{M \times d_k}$
Hopfield Context Retrieval
$Q = XW_q \in \mathbb{R}^{N \times d_k}$
$A_{\text{hop}} = \text{softmax}(\frac{QK^\top}{\sqrt{d_k}}) \in [0,1]^{N \times M}$
$X' = \sigma(\alpha)\, A_{\text{hop}} V + (1 - \sigma(\alpha))\, X \in \mathbb{R}^{N \times d}$
A_{hop}
Context-Enriched Features $X' \in \mathbb{R}^{N \times d}$
Entropy Gate G_{ctx}
$H_i = -\sum_m A_{\text{hop},im} \log A_{\text{hop},im}$
$G_{\text{ctx},i} = e^{-\lambda H_i / \log M}$
Dense Adjacency A
$A = \text{softmax}(\frac{X'X'^\top}{\sqrt{d}}) \in \mathbb{R}^{N \times N}$
Evidential Gate G_{ep}
$(\mu, \nu, \alpha, \beta) = f_{\text{nig}}(\text{sg}(X'))$
$G_{\text{ep},i} = \frac{2\beta_i(1+\nu_i)}{\nu_i(\alpha_i - 1)}$
$\boldsymbol{G} = G_{\text{ctx}} \odot G_{\text{ep}} \in (0,1]^N$
$\tilde{A}_{ij} = A_{ij} \cdot G_i \cdot G_j$
GCN + Residual
$Z = X' + \text{GELU}(\tilde{A}\, X' W_{\text{gcn}})$
NIG Prediction Head
$\hat{y} = \mu, \quad \sigma^2 = \frac{\beta(\nu+1)}{\nu(\alpha-1)}$